

2. Carbon and its compounds are important in the chemical industry.

(a) Carbon can exist in multiple forms. Two of these are fullerenes and diamond.

(i) Name another form of carbon.

1

(ii) Both diamond and the fullerene, C_{60} , can change directly from a solid to a gas. This is called sublimation. For diamond this occurs at $3825\text{ }^{\circ}\text{C}$. The fullerene changes from a solid to a gas at approximately $550\text{ }^{\circ}\text{C}$. Complete the table below to show the strongest type of attraction broken when diamond and the fullerene sublime.

2

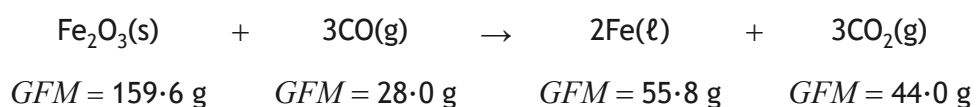
Form of carbon	Strongest attraction broken
diamond	
fullerene	

(iii) The fullerene, C_{60} , reacts with bromine solution in an addition reaction to produce the bromofullerene, $C_{60}Br_{24}$.

Determine the number of double bonds present in a molecule of C_{60} .

1

(b) Carbon can combine with oxygen to make carbon monoxide, CO. Carbon monoxide is used in the production of iron from iron(III) oxide.



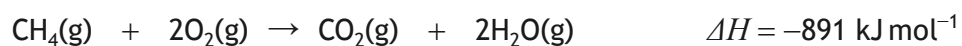
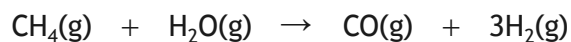
Calculate the atom economy for the production of iron.

2



2. (continued)

- (c) Carbon monoxide can be produced by the reaction of methane and steam.



Calculate the enthalpy change, in kJ mol^{-1} , for this reaction.

2

[Turn over]



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